

Table of Contents

Advanced Graph Problems

1. 3303 — *Nearest Shops (no solution found)*
2. 1134 — *Prüfer Code (no solution found)*
3. 1702 — Tree Traversals
4. 1757 — Course Schedule II
5. 1756 — Acyclic Graph Edges
6. 2177 — Strongly Connected Edges
7. 2179 — Even Outdegree Edges
8. 1707 — Graph Girth
9. 3357 — *Fixed Length Walk Queries (no solution found)*
0. 3111 — Transfer Speeds Sum
1. 3407 — *MST Edge Check (no solution found)*
2. 3408 — *MST Edge Set Check (no solution found)*
3. 3409 — *MST Edge Cost (no solution found)*
4. 1677 — Network Breakdown
5. 3114 — *Tree Coin Collecting I (no solution found)*
6. 3149 — *Tree Coin Collecting II (no solution found)*
7. 1700 — Tree Isomorphism I
8. 1701 — Tree Isomorphism II
9. 1699 — Flight Route Requests
0. 1703 — Critical Cities
1. 1203 — Visiting Cities
2. 3308 — *Graph Coloring (no solution found)*
3. 3158 — *Bus Companies (no solution found)*
4. 3358 — *Split into Two Paths (no solution found)*
5. 1704 — Network Renovation
6. 1705 — *Forbidden Cities (no solution found)*
7. 1752 — Creating Offices
8. 1685 — *New Flight Routes (no solution found)*

CSES Problem Set

Tree Traversals

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are three common ways to traverse the nodes of a binary tree:

- *Preorder*: First process the root, then the left subtree, and finally the right subtree.
- *Inorder*: First process the left subtree, then the root, and finally the right subtree.
- *Postorder*: First process the left subtree, then the right subtree, and finally the root.

There is a binary tree of n nodes with distinct labels. You are given the preorder and inorder traversals of the tree, and your task is to determine its postorder traversal.

Input

The first input line has an integer n : the number of nodes. The nodes are numbered $1, 2, \dots, n$.

After this, there are two lines describing the preorder and inorder traversals of the tree. Both lines consist of n integers.

You can assume that the input corresponds to a binary tree.

Output

Print the postorder traversal of the tree.

Constraints

- $1 \leq n \leq 10^5$

Example

Input:

```
5
5 3 2 1 4
3 5 1 2 4
```

Output:

```
3 1 4 2 5
```

CSES Problem Set**Course Schedule II**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You want to complete n courses that have requirements of the form "course a has to be completed before course b ".

You want to complete course 1 as soon as possible. If there are several ways to do this, you want then to complete course 2 as soon as possible, and so on.

Your task is to determine the order in which you complete the courses.

Input

The first input line has two integers n and m : the number of courses and requirements. The courses are numbered $1, 2, \dots, n$.

Then, there are m lines describing the requirements. Each line has two integers a and b : course a has to be completed before course b .

You can assume that there is at least one valid schedule.

Output

Print one line having n integers: the order in which you complete the courses.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
4 2
2 1
2 3
```

Output:

```
2 1 3 4
```

CSES Problem Set

Acyclic Graph Edges

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given an undirected graph, your task is to choose a direction for each edge so that the resulting directed graph is acyclic.

Input

The first input line has two integers n and m : the number of nodes and edges. The nodes are numbered $1, 2, \dots, n$.

After this, there are m lines describing the edges. Each line has two distinct integers a and b : there is an edge between nodes a and b .

Output

Print m lines describing the directions of the edges. Each line has two integers a and b : there is an edge from node a to node b . You can print any valid solution.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
3 3
1 2
2 3
3 1
```

Output:

```
1 2
3 2
3 1
```

CSES Problem Set**Strongly Connected Edges**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given an undirected graph, your task is to choose a direction for each edge so that the resulting directed graph is strongly connected.

Input

The first input line has two integers n and m : the number of nodes and edges. The nodes are numbered $1, 2, \dots, n$.

After this, there are m lines describing the edges. Each line has two integers a and b : there is an edge between nodes a and b .

You may assume that the graph is simple, i.e., there are at most one edge between two nodes and every edge connects two distinct nodes.

Output

Print m lines describing the directions of the edges. Each line has two integers a and b : there is an edge from node a to node b . You can print any valid solution.

If there are no solutions, only print IMPOSSIBLE.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
3 3
1 2
1 3
2 3
```

Output:

```
1 2
2 3
3 1
```

CSES Problem Set**Even Outdegree Edges**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given an undirected graph, your task is to choose a direction for each edge so that in the resulting directed graph each node has an even outdegree. The outdegree of a node is the number of edges coming out of that node.

Input

The first input line has two integers n and m : the number of nodes and edges. The nodes are numbered $1, 2, \dots, n$.

After this, there are m lines describing the edges. Each line has two integers a and b : there is an edge between nodes a and b .

You may assume that the graph is simple, i.e., there is at most one edge between any two nodes and every edge connects two distinct nodes.

Output

Print m lines describing the directions of the edges. Each line has two integers a and b : there is an edge from node a to node b . You can print any valid solution.

If there are no solutions, only print IMPOSSIBLE.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
4 4
1 2
2 3
3 4
1 4
```

Output:

```
1 2
3 2
3 4
1 4
```

CSES Problem Set

Graph Girth

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given an undirected graph, your task is to determine its *girth*, i.e., the length of its shortest cycle.

Input

The first input line has two integers n and m : the number of nodes and edges. The nodes are numbered $1, 2, \dots, n$.

After this, there are m lines describing the edges. Each line has two integers a and b : there is an edge between nodes a and b .

You may assume that there is at most one edge between each two nodes.

Output

Print one integer: the girth of the graph. If there are no cycles, print -1 .

Constraints

- $1 \leq n \leq 2500$
- $1 \leq m \leq 5000$

Example

Input:

```
5 6
1 2
1 3
2 4
2 5
3 4
4 5
```

Output:

```
3
```

CSES Problem Set

Transfer Speeds Sum

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A computer network has n computers and $n - 1$ connections between two computers. Information can be exchanged between every pair of computers using the connections.

Each connection has a certain transfer speed. Let $d(a, b)$ denote the transfer speed between computers a and b , which is the speed of the slowest connection on the route between a and b . Your task is to compute the sum of transfer speeds between all pairs of computers.

Input

The first line contains the integer n : the number of computers. The computers are numbered $1, 2, \dots, n$.

After this, there are $n - 1$ lines, which describe the connections. Each line has three integers a, b and x : there is a connection between computers a and b with transfer speed x .

Output

Print one integer: the sum of transfer speeds.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x \leq 10^6$

Example

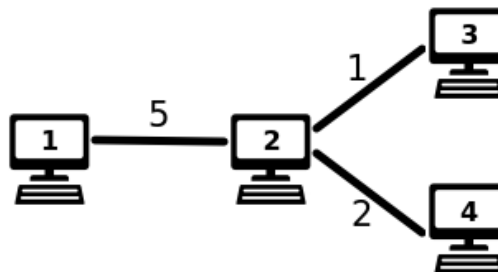
Input:

```
4
1 2 5
2 3 1
2 4 2
```

Output:

```
12
```

Explanation: The following figure corresponds to the sample input:



Here $d(1, 2) = 5$, $d(1, 3) = 1$, $d(1, 4) = 2$, $d(2, 3) = 1$, $d(2, 4) = 2$, and $d(3, 4) = 1$, so the sum of transfer speeds is 12.

CSES Problem Set**Network Breakdown**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Syrjälä's network has n computers and m connections between them. The network consists of components of computers that can send messages to each other.

Nobody in Syrjälä understands how the network works. For this reason, if a connection breaks down, nobody will repair it. In this situation a component may be divided into two components.

Your task is to calculate the number of components after each connection breakdown.

Input

The first input line has three integers n , m and k : the number of computers, connections and breakdowns. The computers are numbered $1, 2, \dots, n$.

Then, there are m lines describing the connections. Each line has two integers a and b : there is a connection between computers a and b . Each connection is between two different computers, and there is at most one connection between two computers.

Finally, there are k lines describing the breakdowns. Each line has two integers a and b : the connection between computers a and b breaks down.

Output

After each breakdown, print the number of components.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq k \leq m$
- $1 \leq a, b \leq n$

Example

Input:

```
5 5 3
1 2
1 3
2 3
3 4
4 5
3 4
2 3
4 5
```

Output:

```
2 2 3
```

CSES Problem Set

Tree Isomorphism I

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given two rooted trees, your task is to find out if they are *isomorphic*, i.e., it is possible to draw them so that they look the same.

Input

The first input line has an integer t : the number of tests. Then, there are t tests described as follows:

The first line has an integer n : the number of nodes in both trees. The nodes are numbered $1, 2, \dots, n$, and node 1 is the root.

Then, there are $n - 1$ lines describing the edges of the first tree, and finally $n - 1$ lines describing the edges of the second tree.

Output

For each test, print "YES", if the trees are isomorphic, and "NO" otherwise.

Constraints

- $1 \leq t \leq 1000$
- $2 \leq n \leq 10^5$
- the sum of all values of n is at most 10^5

Example

Input:

```
2
3
1 2
2 3
1 2
1 3
3
1 2
2 3
1 3
3 2
```

Output:

```
NO
YES
```

CSES Problem Set

Tree Isomorphism II

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given two (not rooted) trees, your task is to find out if they are *isomorphic*, i.e., it is possible to draw them so that they look the same.

Input

The first input line has an integer t : the number of tests. Then, there are t tests described as follows:

The first line has an integer n : the number of nodes in both trees. The nodes are numbered $1, 2, \dots, n$.

Then, there are $n - 1$ lines describing the edges of the first tree, and finally $n - 1$ lines describing the edges of the second tree.

Output

For each test, print "YES", if the trees are isomorphic, and "NO" otherwise.

Constraints

- $1 \leq t \leq 1000$
- $2 \leq n \leq 10^5$
- the sum of all values of n is at most 10^5

Example

Input:

```
2
3
1 2
2 3
1 2
1 3
3
1 2
2 3
1 3
3 2
```

Output:

```
YES
YES
```

CSES Problem Set**Flight Route Requests**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n cities with airports but no flight connections. You are given m requests which routes should be possible to travel.

Your task is to determine the minimum number of one-way flight connections which makes it possible to fulfil all requests.

Input

The first input line has two integers n and m : the number of cities and requests. The cities are numbered $1, 2, \dots, n$.

After this, there are m lines describing the requests. Each line has two integers a and b : there has to be a route from city a to city b . Each request is unique.

Output

Print one integer: the minimum number of flight connections.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
4 5
1 2
2 3
2 4
3 1
3 4
```

Output:

```
4
```

Explanation: You can create the connections $1 \rightarrow 2$, $2 \rightarrow 3$, $2 \rightarrow 4$ and $3 \rightarrow 1$. Then you can also fly from city 3 to city 4 using the route $3 \rightarrow 1 \rightarrow 2 \rightarrow 4$.

CSES Problem Set**Critical Cities**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n cities and m flight connections between them. A city is called a *critical city* if it appears on every route from a city to another city.

Your task is to find all critical cities from Syrjälä to Lehmälä.

Input

The first input line has two integers n and m : the number of cities and flights. The cities are numbered $1, 2, \dots, n$. City 1 is Syrjälä, and city n is Lehmälä.

Then, there are m lines describing the connections. Each line has two integers a and b : there is a flight from city a to city b . All flights are one-way.

You may assume that there is a route from Syrjälä to Lehmälä.

Output

First print an integer k : the number of critical cities. After this, print k integers: the critical cities in increasing order.

Constraints

- $2 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5 5
1 2
2 3
2 4
3 5
4 5
```

Output:

```
3
1 2 5
```

CSES Problem Set**Visiting Cities**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You want to travel from Syrjälä to Lehmälä by plane using a minimum-price route. Which cities will you certainly visit?

Input

The first input line contains two integers n and m : the number of cities and the number of flights. The cities are numbered $1, 2, \dots, n$. City 1 is Syrjälä, and city n is Lehmälä.

After this, there are m lines describing the flights. Each line has three integers a , b , and c : there is a flight from city a to city b with price c . All flights are one-way flights.

You may assume that there is a route from Syrjälä to Lehmälä.

Output

First print an integer k : the number of cities that are certainly in the route. After this, print the k cities sorted in increasing order.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$
- $1 \leq c \leq 10^9$

Example

Input:

```
5 6
1 2 3
1 3 4
2 3 1
2 4 5
3 4 1
4 5 8
```

Output:

```
4
1 3 4 5
```

CSES Problem Set

Network Renovation

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Syrjälä's network consists of n computers and $n - 1$ connections between them. It is possible to send data between any two computers.

However, if any connection breaks down, it will no longer be possible to send data between some computers. Your task is to add the minimum number of new connections in such a way that you can still send data between any two computers even if any single connection breaks down.

Input

The first input line has an integer n : the number of computers. The computers are numbered $1, 2, \dots, n$.

After this, there are $n - 1$ lines describing the connections. Each line has two integers a and b : there is a connection between computers a and b .

Output

First print an integer k : the minimum number of new connections. After this, print k lines describing the connections. You can print any valid solution.

Constraints

- $3 \leq n \leq 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5
1 2
1 3
3 4
3 5
```

Output:

```
2
2 4
4 5
```

CSES Problem Set**Creating Offices**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n cities and $n - 1$ roads between them. There is a unique route between any two cities, and their distance is the number of roads on that route.

A company wants to have offices in some cities, but the distance between any two offices has to be at least d . What is the maximum number of offices they can have?

Input

The first input line has two integers n and d : the number of cities and the minimum distance. The cities are numbered $1, 2, \dots, n$.

After this, there are $n - 1$ lines describing the roads. Each line has two integers a and b : there is a road between cities a and b .

Output

First print an integer k : the maximum number of offices. After that, print the cities which will have offices. You can print any valid solution.

Constraints

- $1 \leq n, d \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5 3
1 2
2 3
3 4
3 5
```

Output:

```
2
1 4
```